

SWARNALI MOLLICK

DATA SCIENTIST | SPECIALISING IN APPLIED AI

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LinkedIn: [linkedin.com/in/s-mollick](https://www.linkedin.com/in/s-mollick) | **GitHub:** github.com/itsmeSwarnali | **Google Scholar:** [Scholar Profile](#)

Portfolio: swarnalimollick.vercel.app

PROFESSIONAL SUMMARY

AI and Data Engineering MSc candidate with hands-on experience building end-to-end Data Science and Applied AI systems. Skilled in Python, SQL, statistical analysis, machine learning, NLP, RAG, API development and deployment. Delivered classification, forecasting and transformer-based applications using Scikit-learn, PyTorch, Hugging Face, FastAPI, Docker and Streamlit. Co-author of 10+ peer-reviewed publications. Seeking a Data Scientist opportunity where I can contribute to machine-learning and applied-AI projects.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, Pandas, NumPy, PySpark, data cleaning, EDA, feature engineering, statistical analysis

Machine Learning: Scikit-learn, classification, regression, forecasting, cross-validation, imbalanced-data analysis, model evaluation

Deep Learning & NLP: PyTorch, TensorFlow/Keras, Transformers, DistilBERT, Hugging Face, LangChain, RAG, embeddings, ChromaDB, prompt engineering, semantic search

Deployment & Tools: FastAPI, Docker, Streamlit, Flask, AWS fundamentals, Git/GitHub, Jupyter Notebook, VS Code, APIs, technical documentation

Visualization: Matplotlib, Seaborn, Plotly, data storytelling, business insights

SELECTED DATA SCIENCE & AI PROJECTS

AI/NLP Learning Assistant - Retrieval-Augmented Generation | [GitHub](#)

Live demo: ai-nlp-learning-assistant.streamlit.app

- Built a RAG assistant using LangChain, ChromaDB, Hugging Face embeddings and Llama 3.1 through the Groq inference API to retrieve learning materials and generate context-grounded responses.
- Integrated intent classification, document ingestion, semantic search and conversation history; deployed the application on Streamlit Cloud.

Financial Sentiment Classification & Streamlit App | [GitHub](#)

Live demo: financial-sentiment.streamlit.app

- Designed a three-class financial NLP experiment comparing fine-tuned DistilBERT with few-shot Llama 3.1 prompting for Bearish, Bullish and Neutral text classification.
- DistilBERT achieved 85.8% accuracy and 82.0% Macro-F1, outperforming few-shot Llama 3.1 in accuracy and inference latency; deployed the selected model as a Streamlit application.

Customer Churn Prediction - Machine Learning API | [GitHub](#)

- Built an end-to-end churn pipeline for 7,043 telecom customers, covering EDA, preprocessing, feature engineering and comparison of Logistic Regression, Random Forest and XGBoost.
- Achieved ROC-AUC 0.8409 and 80.41% accuracy with Logistic Regression, translated the findings into retention recommendations, and deployed the pipeline using FastAPI and Docker.

Retail Sales Prediction and Business Analytics | [GitHub](#)

- Analyzed 1,017,209 Rossmann retail records to quantify the effects of promotions, holidays, seasonality and store characteristics; promotions increased average sales by approximately 39%.
- Compared Linear Regression, Decision Tree, Gradient Boosting, XGBoost and Random Forest; the best model achieved R^2 0.868, MAE 740.34 and RMSE 1,128.59.

PROFESSIONAL EXPERIENCE

Web Developer - Appstick, Bangladesh

Sep 2022 - Sep 2023

- Developed and maintained responsive, database-backed web applications using HTML, CSS, Bootstrap, Laravel, PHP and MySQL.
- Implemented MySQL-driven features, maintained structured data flow across frontend, backend and database layers, and documented development workflows.

Web Designer Intern - The Virtual BD, Bangladesh

Apr 2021 - Feb 2022

- Designed responsive WordPress and Elementor pages, implemented HTML-based customizations and received the Best Web Designer Award for internship performance.

EDUCATION

MSc in Computer Science and Information Technology

Sep 2024 - Expected Dec 2026

Major: Artificial Intelligence and Data Engineering

Ca' Foscari University of Venice, Italy

Coursework completed; Master's thesis in progress | Current weighted average: 27.13/30

- Relevant coursework: Foundations of Machine Learning, Foundations of Artificial Intelligence, Advanced Data Management, Calculus and Optimization, Image and Video Understanding, Deep Learning for NLP, Time Series for Computer Science.
- Master's thesis in progress: Financial market prediction using graph neural networks and LLMs.

BSc in Computer Science and Engineering

2018 - 2022

Northern University of Business and Technology Khulna, Bangladesh | CGPA 3.959/4.00 | 152 credits

- Undergraduate thesis: *Adapted Deep Ensemble Learning-Based Voting Classifier for Osteosarcoma Cancer Classification*.

RESEARCH & SELECTED PUBLICATIONS

Co-author of 10+ journal and conference publications in machine learning, healthcare AI, data mining, and environmental prediction. Full record: [Google Scholar](#)

- Adapted Deep Ensemble Learning-Based Voting Classifier for Osteosarcoma Cancer Classification — Diagnostics, 2023.
- A Comparative Study, Prediction and Development of Chronic Kidney Disease Using Machine Learning on Patient Clinical Records — Human-Centric Intelligent Systems, 2023.
- Efficient Prediction of Water Quality Index Using Machine-Learning Algorithms — Human-Centric Intelligent Systems, 2021 — Best Paper Award.

HONOURS & AWARDS

- Italian Government MAECI Scholarship, 2024-2026.
- Best Paper Award, Human-Centric Intelligent Systems, Springer, 2022.
- Best Web Designer Award, The Virtual BD, 2021.
- NUBTK Departmental Scholarship for Excellent Result, 2018 and 2019.

COURSES & CERTIFICATIONS

IBM/ Coursera: Machine Learning with Python | **aiQuest:** Data Science and Machine Learning with Python | **Ostad:** Data Science Certificate Program | **MathWorks:** MATLAB Onramp

ACADEMIC SERVICE, LEADERSHIP & LANGUAGES

Academic service: Conference Reviewer, International Conference on Computer Science and Application Engineering (CSAE), 2023; IEEE Student Member, 2022-2023.

Leadership: Class Representative, NUBTK; Programming Club member who organized a programming competition and delivered AI-focused talks to junior students.

Languages: Bengali (native), English (C1), Italian (A1, actively learning), Hindi (conversational).